DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Automata Theory and Computability | Course Code: 21CS51 | Credits: 4

Exam Session: End Semester Exam 2024 **Max Marks: 80 | Duration: 3 Hours**

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. State and prove the Pumping Lemma for Regular Languages. Use it to prove that the language L = { a^n b^n | n >= 1 } is NOT regular. **[8 Marks]**

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Q2. Design a Pushdown Automata (PDA) to accept the language L = { w c w^R | w in {a, b}* } by empty store (null stack). Provide state transition table and instant description for string "ab c ba". **[8 Marks]**